



Description of Variable Clas

Clas bit : preclassification of event. Example:
Clas 16 = hadronic event

Description of Variable Cthr

Cosine of the thrust axis polar angle (angle from the beam (+z) to the thrust axis), [-1,1].
Calculated from good charged tracks (see standard ALEPH definition of good charged tracks in description of variable **ngood**) If there are only two good charged tracks, it is the cosine of the positive track.

Description of Variable D0

minimum distance of track to interaction point in plane transverse to beam axis (cm). Vector of length **nch** (number of charged tracks)

Description of Variable Ech

Energy sum (GeV) of all charged tracks (no selection)

Description of Variable Eec

Energy sum (GeV) of all objects in ECAL (no selection). The energy of each object is corrected for geometrical effects.

Description of Variable Eecal

energy of the ECAL object (GeV). vector of length **nec** (number of ECAL objects)

Description of Variable Egood

sum of energy (GeV) of good charged tracks, see variable **ngood**

Description of Variable Ehc

Energy sum (GeV) of all objects in HCAL (no selection). The energy of each object is corrected for geometrical effects.

Description of Variable Ehcal

energy of the HCAL object (GeV). vector of length **nhc** (number of HCAL objects)

Description of Variable Elep

centre-of-mass energy (GeV)

Description of Variable Evt

Event number

Description of Variable Nch

Number of all charged tracks (no selection)

Description of Variable Nec

Number of all objects in ECAL (no selection)

Description of Variable Nefl

Number of energy flow objects

Description of Variable Ngood

Number of good charged tracks. according to standard ALEPH selection: $N_{tpc} > 4$, $abs(cos\theta) < 0.95$, $d_0 < 2.$, $z_0 < 20$.

Description of Variable Nhc

Number of all objects in HCAL (no selection)

Description of Variable Nhits

number of hits of the track in the tracking system (TPC in ALEPH). Vector of length **nch** (number of charged tracks)

Description of Variable Njet

number of jets found [0,10]. Jets found by Durham E-scheme, $Y_{cut} = 0.04$; jets are found with "good" charged tracks (see standard ALEPH definition in description of variable **ngood**)

Description of Variable Nplanes

number of hits in the last planes of HCAL (for muon identification) [0,11] Vector of length **nch** (number of charged tracks)

Example: `nplanes[1]` gives the number of hits of the second charged track of each event.

Description of Variable pcha

`pcha(3)`: Track momentum (px,py,pz) (GeV/c). Vector of length **nch***3 (number of charged tracks)

Example: `pcha[2][3]` gives the third (z) component of the second charged track of each event.

Description of Variable Pecal

Phi of the ECAL object (radians) [0, 2], vector of length **nec** (number of ECAL objects)

Description of Variable pefl

pefl(4): Four momentum (px,py,pz,E) (GeV/c).
Vector of length **nefl***4 (number of energy flow objects)

Description of Variable Phcal

Phi of the HCAL object (radians) [0, 2], vector of length **nhc** (number of ECAL objects)

Description of Variable pjet

pjet(4): jet four momentum (px,py,pz,E) (GeV).
vector of length **njet***4

Description of Variable Qcha

Charge of the track [-1 or 1], vector of length **nch***2 (number of charged tracks)

Description of Variable Run

Run number. The runs are selected to be good runs where all the subdetectors of the apparatus work well.

Description of Variable Tecal

Theta of the ECAL object (radians) [0,1], vector of length **nec** (number of ECAL objects)

Description of Variable Thcal

Theta of the HCAL object (radians) [0,1], vector of length **nhc** (number of HCAL objects)

Description of Variable Thruv

Thrustvalue

Description of Variable Typeefl

Type of energy flow object [1-10] 1-3:charged track, 4:electromagnetic object (ECAL),photon 5:neutral hadron (HCAL) 7:LCAL 8:SICAL (luminometers) Vector of length **nefl** (number of energy flow objects)

Description of Variable Z0

minimum distance of track to interaction point along beam axis (cm). Vector of length **nch** (number of charged tracks)